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GREATER NOIDA, UTTAR PRADESH

SUNSHINE READYMIX — READY-MIX CONCRETE (RMC) PLANT

Research & Development in Concrete Production

Internship Presentation & Project Report

INTERNSHIP PERIOD

June 2026 – August 2026

STUDENT CREDENTIALS

Harshit Parmar

Enrollment / Roll No.

2410031422

Program & Semester

**B.Tech (CSE)
5th Semester**

ORGANIZATION

Sunshine Readymix

Research & Development (R&D) Department

Candidate & Program Overview

Student Profile

Name: Harshit Parmar
Roll Number: 2410031422
Degree: B.Tech (Computer Science & Engineering)
Academic Batch: 2024–2028
University: IILM University, Greater Noida, U.P.

Internship Profile

Organization: Sunshine Readymix
Department: Research & Development (R&D)
Industry: Ready-Mix Concrete (RMC)
Duration: 1 June 2026 – 31 August 2026
Certificate Issued: 1 September 2026

Project Theme

Hands-on industrial exposure to concrete technology, mix design, quality control, material testing, and ready-mix concrete production processes.

The internship bridges academic concepts with practical, industry-oriented concrete production.

Organization Profile

Sunshine Readymix and the Ready-Mix Concrete industry context

Sunshine Readymix

A Ready-Mix Concrete producer operating a batching plant at Rithoj Road, Village Bhondsi, Haryana.

R&D Department

Responsible for mix design development, raw material validation, and batch-to-batch consistency.

Industry Relevance

Construction projects need concrete delivered at a specified strength, workability, and durability.

Learning Environment

A live production plant and testing laboratory provided a structured environment for applied learning.

Core Focus

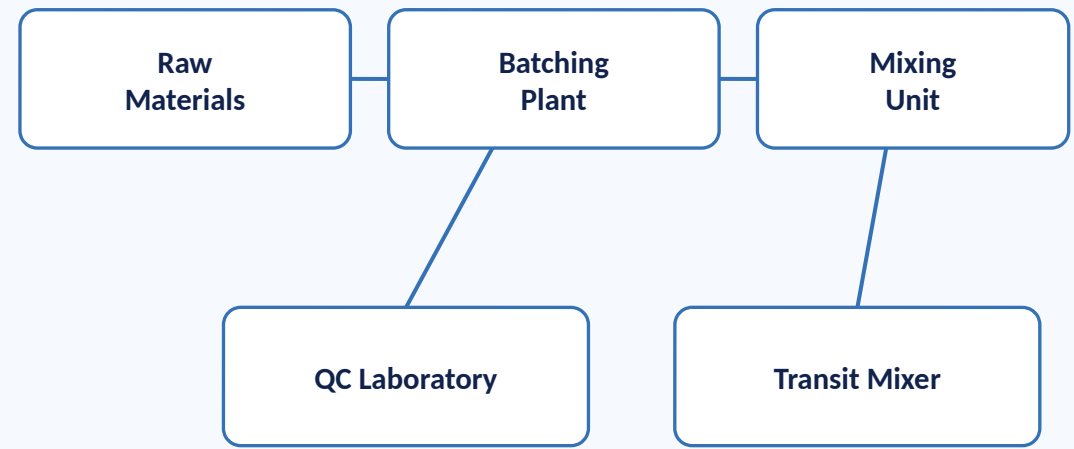
Concrete technology, mix design, quality control, material testing, and production processes.

Problem Statement & Industry Context

The Industry Challenge

- Concrete must reach site at a specified strength, workability, and durability.
- Raw materials, ambient conditions, and transit time vary continuously.
- Every batch must be proportioned, weighed, and mixed under tight control.
- Fresh and hardened concrete must be verified before and after dispatch.

Ready-Mix Concrete Production Flow



Objective: understand how a plant delivers consistent, specification-compliant concrete through mix design, testing, and quality control.

Project Objectives

01 Concrete Technology

Understand cement, aggregates, water, and admixtures, and how each affects the final product.

02 Mix Design

Study the proportioning of concrete mixes for target grades and the adjustment of trial mixes.

03 Quality Control

Observe the quality assurance procedures applied at every stage of production.

04 Material Testing

Perform standard laboratory tests on raw materials and on fresh and hardened concrete.

05 Production Processes

Follow the end-to-end RMC plant operation from storage and batching to delivery.

Materials, Tools & Standards Used

Materials & Constituents

- Cement (OPC / PPC)
- Coarse and fine aggregates
- Potable water
- Chemical admixtures

Mix Design & Standards

- Mix proportioning guidelines
- Target mean strength
- Water-cement ratio
- Trial mix adjustment

Laboratory & Plant

- Compression testing machine
- Slump cone apparatus
- Sieve shaker & balances
- Computerised batching plant

Technical Focus

Concrete Technology

Mix Design

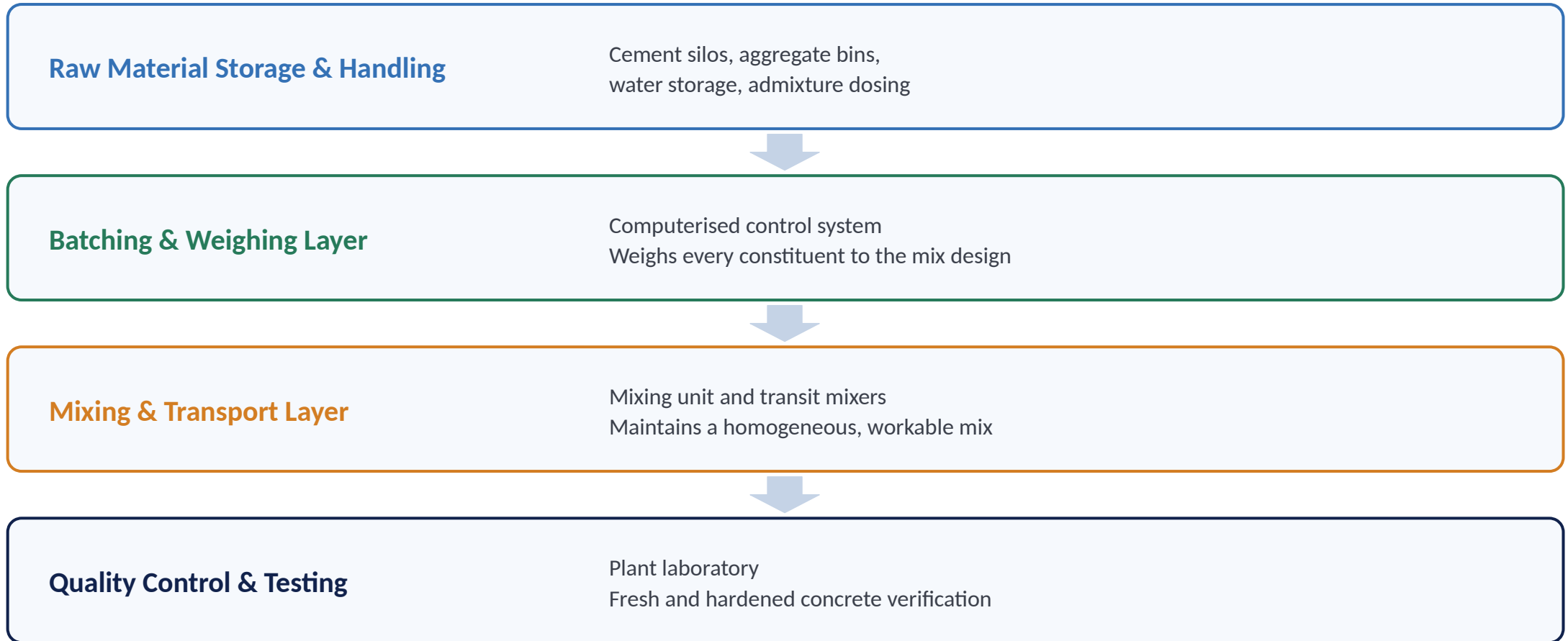
Quality Control

Material Testing

Production Processes

System Architecture

Multi-stage plant architecture designed for separation of concerns



Separation of stages enables clearer material flow, easier testing, and traceable quality control.

Methodology & Execution

Sequential, progressive execution across the three months of the internship.



Outcome: laboratory understanding was tied directly to live plant production and quality records.

Project Scope & Deliverables

Included in Scope

- Raw material receipt and testing
- Concrete mix design and proportioning
- Computerised batching and mixing
- Fresh concrete testing at discharge
- Compressive strength verification
- Quality control documentation
- RMC plant production processes

Deliberately Outside Scope

- Structural design of concrete members
- Site placement and compaction
- Formwork and shuttering
- Commercial and logistics planning

This maintained a strict focus on core concrete production and quality control.

The scope covered the full batch lifecycle, from raw material testing through to verified delivery.

Expected Outcomes & Skills Demonstrated

- 1 Material Knowledge** Understanding of cement, aggregates, water, and admixtures and their influence on concrete.
- 2 Mix Design Competence** Ability to follow the proportioning procedure for a target grade and adjust trial mixes.
- 3 Testing Proficiency** Practical competence in workability measurement and compressive strength verification.
- 4 Process Understanding** Complete view of the RMC plant from storage and batching through mixing and delivery.
- 5 Quality Assurance** Appreciation of the records that make every production batch traceable and auditable.

Overall result: practical, industry-grounded competence in ready-mix concrete production and quality control.

Internship Completion Certificate

Certificate issued by Sunshine Readymix for the R&D internship



GSTIN: 06CNC PG9734Q2ZD

INTERSHIP CERTIFICATE

01st Sep 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. Harshit Parmar has successfully completed a three-month internship in the Research & Development (R&D) Department at our Ready-Mix Concrete (RMC) Plant from 1 June 2026 to 31 August 2026.

During his internship, he was involved in activities related to concrete technology, mix design, quality control, material testing, production processes, and research & development of concrete mixes. He gained practical exposure to RMC plant operations and developed an understanding of the various testing and quality parameters associated with concrete production.

His conduct and performance during the internship were found to be satisfactory. He demonstrated sincerity, enthusiasm, and a willingness to learn throughout the internship period.

We wish Mr. Harshit Parmar all the very best for his future academic and professional endeavors.

Internship Duration: 1 June 2026 – 31 August 2026
Department: Research & Development (R&D)
Industry: Ready-Mix Concrete (RMC)

For Sunshine Readymix



Authorized Signatory

Certificate of Completion — 1 September 2026

References

- Sunshine Readymix, “Internship Certificate,” R&D Department, Ready-Mix Concrete Plant, 1 September 2026.
- Bureau of Indian Standards, “IS 456 : 2000 — Plain and Reinforced Concrete — Code of Practice.”
- Bureau of Indian Standards, “IS 10262 : 2019 — Concrete Mix Proportioning — Guidelines.”
- Bureau of Indian Standards, “IS 383 : 2016 — Coarse and Fine Aggregate for Concrete — Specification.”
- Bureau of Indian Standards, “IS 1199 : 2018 — Fresh Concrete — Methods of Sampling, Testing and Analysis.”
- Bureau of Indian Standards, “IS 516 : 2021 — Hardened Concrete — Methods of Test.”

Source Basis

Internship details and completion information are grounded in the certificate issued by Sunshine Readymix.

Thank You!

Harshit Parmar

Roll No: 2410031422

B.Tech (CSE) | 5th Semester

School of Computer Science & Engineering

IILM University, Greater Noida, U.P.